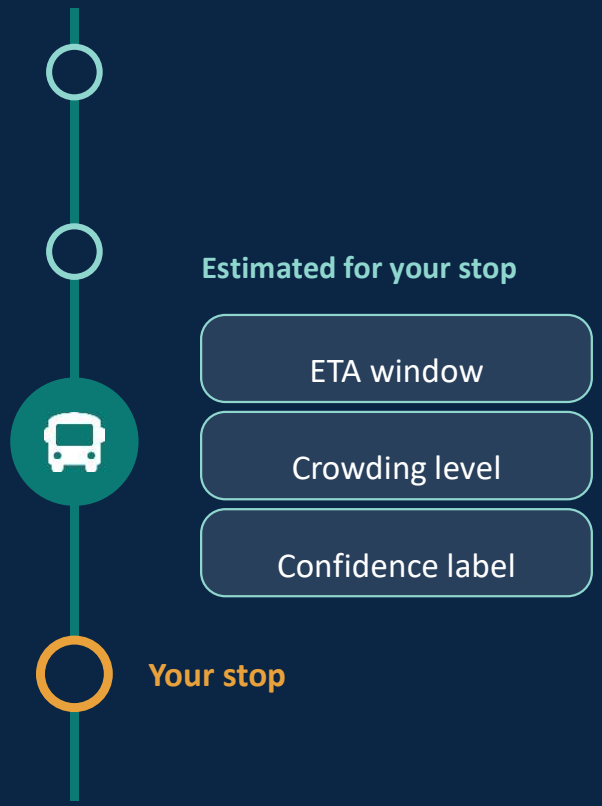
SeatSignal

Know when your bus arrives — and how full it may be.

A concept for a stop-specific bus information tool. It would combine an arrival estimate, a crowding estimate and a plain-language confidence label.



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The problem: deciding without the full picture

1. Before leaving, a rider wants two answers: when will my bus reach my stop, and will there be space?
2. Scheduled times can be rough. Delhi's open timetable data estimates stop times at a constant speed.
3. Live positions exist, but the feed needs a registered API key, and not every route updates live.
4. Knowing how full a bus will be is harder to find than knowing when it arrives.
5. Missing, to be validated: one stop-level answer covering timing, crowding and how reliable both are.

When will it arrive?
Arrival estimate for my stop

Will there be space?
Crowding level of that bus

How sure is the estimate?
The piece we want to test

Who is affected, and why it matters

Daily commuters and students

People who plan a trip around one stop and one bus. A wrong guess costs waiting time.

Riders who need space

Older riders, parents with children, people with luggage or mobility needs may care most about room on board.

Transport agencies

Agencies could use crowding information to improve what they tell riders.

Why it matters?

Seat and space availability is part of bus service quality and shapes how riders feel about the bus.

Possible impact (to be tested, not claimed)

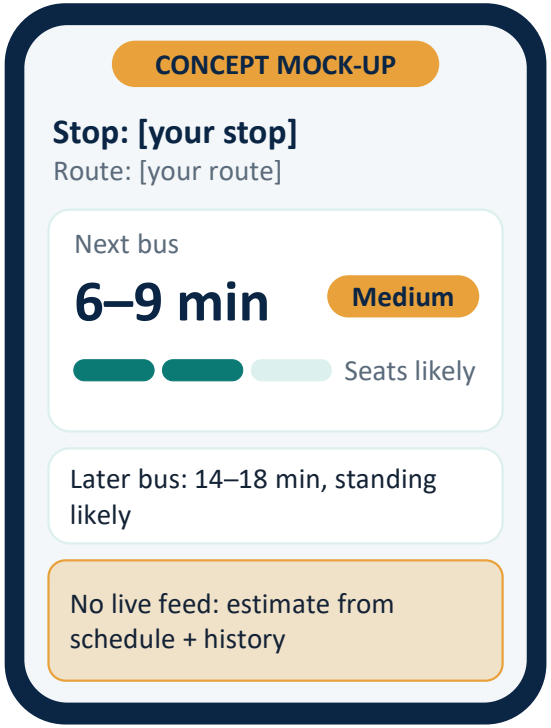
If solved: less guessing, better choice of when to leave. If not: riders keep guessing, with avoidable waiting and less trust in bus information.

SeatSignal: the proposed solution (concept)

1. Rider selects a bus stop and route
2. System gathers live position (if available), schedule and past trips
3. Models estimate an arrival window and a crowding level
4. App shows both, with a confidence label
5. If live data is missing, show a clearly marked fallback estimate

HYPOTHESIS

A combined view with a confidence label helps riders decide better than an arrival time alone.



Where ML fits — and where it doesn't



Ordinary software

- Pick stop and route
- Read the bus schedule
- Fetch live bus positions
- Count stops remaining
- Flag stale or missing feeds
- Show results on screen

Where ML may help



Arrival time (regression)
Learn the gap between scheduled and actual time from past trips, using position, stops left, time of day, recent delay and weather.



Crowding level (classification)
Learn typical load by route, stop and time, then update it with loads at earlier stops.



Confidence
Output a range or class probabilities instead of one number, so the app can say how sure it is.

ML approach and data requirements

Likely ML problem types

Regression
Arrival time in minutes: a continuous number.

Time-series forecasting
Delay and crowding patterns by hour and day.

Classification
Crowding level, such as seats available or standing.

Uncertainty estimate (other)
A range or probability behind each answer.

Not needed for v1: computer vision, NLP, generative AI, reinforcement learning. Not sure yet: whether deep models beat simple ones.

Data needed

Bus GPS + timestamps — API KEY
Delhi open transit data live feed; needs an API key

Routes, stops, schedule — PUBLIC
Public download; scheduled times are rough estimates

Actual arrival times — COLLECT
Not provided directly; derive by logging live positions over time

Passenger counts / labels — PERMISSION
Not in OTD pages we read; needs operator data or hand labels

Traffic, weather, time of day — UNVERIFIED
Time is free; weather and traffic sources not yet checked

Data types: tabular numbers + location/time data.

Challenges, feasibility and next steps

What could go wrong

Data availability and coverage
Feeds may be patchy; only covered routes work.

Data quality
Real bus data often has gaps and duplicate records.

Crowding labels
Counts likely need operator data or hand labels.

Prediction errors
A wrong “seats likely” costs trust, so show ranges.

Privacy and trust
Count people in aggregate only; never track individuals.

Realistic first steps and how to judge success

1. Check access: request an OTD key and log live positions for 1–2 routes.
2. Arrival test: compare our arrival error with the schedule alone.
3. Crowding pilot: hand-label rides on one route and compare with a historical average.
4. Rider check: short interviews on whether the confidence label helps.

Success = beating simple baselines, not hitting a fixed target.

Main uncertainty: can we get reliable crowding labels for even one route? If not, the project shrinks to arrival time plus confidence.

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